

Problem 3

Part 3a)

Part (i):

Since each $X_j : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is a random variable and each $g_j : (\mathbb{R}, \mathcal{B}(\mathbb{R})) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is Borel-measurable, the composition $g_j(X_j)$ is a random variable, i.e., a measurable function from $(\Omega, \mathcal{F})$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. To show that the family $(g_i(X_i))_{i \in I}$ is independent, we need to verify that for any finite subset of indices $J \subset I$ and any choice of Borel sets $B_j \in \mathcal{B}(\mathbb{R})$ for $j \in J$, the events $\{g_j(X_j) \in B_j\} = \{\omega \in \Omega : g_j(X_j(\omega)) \in B_j\}$ are independent.

Let $J \subset I$ such that J is finite and let $B_j \in \mathcal{B}(\mathbb{R}), \forall j \in J$. Since the corresponding g_j is Borel-measurable, then $g_j^{-1}(B_j) \in \mathcal{B}(\mathbb{R})$. Then:

$$\mathbb{P} \left(\bigcap_{j \in J} \{g_j(X_j) \in B_j\} \right) = \mathbb{P} \left(\bigcap_{j \in J} \{X_j \in g_j^{-1}(B_j)\} \right)$$

Since $(X_i)_{i \in I}$ are independent, then by Definition 3.2.5:

$$\begin{aligned} \mathbb{P} \left(\bigcap_{j \in J} \{X_j \in g_j^{-1}(B_j)\} \right) &= \prod_{j \in J} \mathbb{P}(X_j \in g_j^{-1}(B_j)) \\ &= \prod_{j \in J} \mathbb{P}(g_j(X_j) \in B_j) \end{aligned}$$

Since the finite subset $J \subset I$ and the Borel sets B_j were chosen arbitrarily, the family $(g_i(X_i))_{i \in I}$ is independent (Definition 3.2.5). □

Part (ii):

We need to show that $Y = f(X_1, X_2)$ and $Z = g(X_3)$ are independent, i.e., for all $A, B \in \mathcal{B}(\mathbb{R})$:

$$\mathbb{P}(Y \in A, Z \in B) = \mathbb{P}(Y \in A) \cdot \mathbb{P}(Z \in B).$$

Step 1. We first show that (X_1, X_2) and X_3 are independent, i.e., for all $C \in \mathcal{B}(\mathbb{R}^2)$ and $D \in \mathcal{B}(\mathbb{R})$:

$$\mathbb{P}((X_1, X_2) \in C, X_3 \in D) = \mathbb{P}((X_1, X_2) \in C) \cdot \mathbb{P}(X_3 \in D).$$

The family of rectangles $\mathcal{M} := \{B_1 \times B_2 : B_1, B_2 \in \mathcal{B}(\mathbb{R})\}$ is an intersection-stable generator of $\mathcal{B}(\mathbb{R}^2)$ (Definition 2.3.1). By Lemma 3.1.11, the pre-image $\tilde{\mathcal{M}} := \{(X_1, X_2)^{-1}(C) : C \in \mathcal{M}\}$ is an intersection-stable generator of $\sigma(X_1, X_2)$ and, by Lemma 2.2.11 it suffices to verify that $\forall B_1 \times B_2 \in \mathcal{M}$ and $\forall D \in \mathcal{B}(\mathbb{R})$:

$$\mathbb{P}((X_1, X_2) \in B_1 \times B_2, X_3 \in D) = \mathbb{P}((X_1, X_2) \in B_1 \times B_2) \cdot \mathbb{P}(X_3 \in D).$$

The equality above holds because of the independence of X_1, X_2, X_3 :

$$\begin{aligned} \mathbb{P}((X_1, X_2) \in B_1 \times B_2, X_3 \in D) &= \mathbb{P}(X_1 \in B_1, X_2 \in B_2, X_3 \in D) \\ &= \mathbb{P}(X_1 \in B_1) \cdot \mathbb{P}(X_2 \in B_2) \cdot \mathbb{P}(X_3 \in D) \\ &= \mathbb{P}((X_1, X_2) \in B_1 \times B_2) \cdot \mathbb{P}(X_3 \in D). \end{aligned}$$

Step 2. By Lemma 3.1.9, $Y = f(X_1, X_2)$ and $Z = g(X_3)$ are well-defined random variables. Let $A, B \in \mathcal{B}(\mathbb{R})$. Since f and g are Borel-measurable, $f^{-1}(A) \in \mathcal{B}(\mathbb{R}^2)$ and $g^{-1}(B) \in \mathcal{B}(\mathbb{R})$.

Using Step 1:

$$\begin{aligned}\mathbb{P}(Y \in A, Z \in B) &= \mathbb{P}((X_1, X_2) \in f^{-1}(A), X_3 \in g^{-1}(B)) \\ &= \mathbb{P}((X_1, X_2) \in f^{-1}(A)) \cdot \mathbb{P}(X_3 \in g^{-1}(B)) \\ &= \mathbb{P}(Y \in A) \cdot \mathbb{P}(Z \in B).\end{aligned}$$

Since A, B were arbitrary, Y and Z are independent by Definition 3.2.1. □

Part 3b)

Since X is $\mathcal{G}$ - $\mathcal{B}(\mathbb{R})$ -measurable, then $X^{-1}(B) \in \mathcal{G}, \forall B \in \mathcal{B}(\mathbb{R})$ and $\sigma(X) \subseteq \mathcal{G}$ (Definition 3.1.10). Since X is $\mathcal{H}$ - $\mathcal{B}(\mathbb{R})$ -measurable, then $X^{-1}(B) \in \mathcal{H}, \forall B \in \mathcal{B}(\mathbb{R})$ and $\sigma(X) \subseteq \mathcal{H}$. Therefore, $\sigma(X) \subseteq \mathcal{G} \cap \mathcal{H}$. In particular, for every $x \in \mathbb{R}$, the event $\{X \leq x\} = X^{-1}((-\infty, x])$ belongs to $\sigma(X)$, and hence $\{X \leq x\} \in \mathcal{G} \cap \mathcal{H}$.

Fix any $x \in \mathbb{R}$ and set $A := \{X \leq x\} \in \mathcal{G} \cap \mathcal{H}$. Since $\mathcal{G}$ and $\mathcal{H}$ are independent σ -algebras, the events $A \in \mathcal{G}$ and $A \in \mathcal{H}$ are independent, so:

$$\mathbb{P}(A \cap A) = \mathbb{P}(A) \cdot \mathbb{P}(A) = (\mathbb{P}(A))^2.$$

Since $A \cap A = A$, and $\mathbb{P}(A) \cdot \mathbb{P}(A) = \mathbb{P}(A)^2$, then:

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A)^2 \\ \iff \mathbb{P}(A) - \mathbb{P}(A)^2 &= 0 \\ \iff \mathbb{P}(A)(1 - \mathbb{P}(A)) &= 0.\end{aligned}$$

The only way for $\mathbb{P}(A)(1 - \mathbb{P}(A))$ to be zero is if $\mathbb{P}(A) = 0$ or $\mathbb{P}(A) = 1$, i.e. if $\mathbb{P}(A) \in \{0, 1\}$. Since $x \in \mathbb{R}$ was arbitrary, then:

$$F_X(x) = \mathbb{P}(X \leq x) \in \{0, 1\} \quad \text{for all } x \in \mathbb{R}.$$

By the fundamental properties of distribution functions, F_X is non-decreasing, right-continuous, and satisfies $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow \infty} F_X(x) = 1$. In particular, there exist $x_0, x_1 \in \mathbb{R}$ such that $F_X(x_0) = 0$ and $F_X(x_1) = 1$, so the following infimum is finite:

$$c := \inf\{x \in \mathbb{R} : F_X(x) = 1\} \in [x_0, x_1] \subset \mathbb{R}.$$

By right-continuity, $F_X(c) = 1$, and for any $\epsilon > 0$, $F_X(c - \epsilon) = 0$. By the definition of the infimum and the fact that $F_X(x) \in \{0, 1\}$, we know $F_X(x) = 0$ for all $x < c$, meaning the left limit is $F_X(c-) = 0$. Consequently, by Lemma 1.4.2:

$$\mathbb{P}(X = c) = F_X(c) - F_X(c-) = 1 - 0 = 1.$$

Thus, X is almost surely constant. □